1. Title

Active Mapping using 3D Scene Graphs: A Review and Implementation

2. Description

This project reviews and implements active mapping methods using hierarchical 3D scene graphs. This is a new and rapidly evolving area of research, focusing on how scene-graph information can guide robot exploration and Next-Best-View (NBV) selection. The project will reproduce selected methods in simulation and, where feasible, evaluate them on TIAGo or B2 robots in the lab. It will compare geometric and semantic exploration strategies, assess mapping and exploration performance, and identify open research challenges in active 3D scene-graph mapping.

3. Methods and Approaches

Task 1: Baseline Framework Programming Analysis and Simulation Reproduction of Active 3D Scene-Graph Mapping

- Compare the evolution from Kimera and Hydra to ConceptGraphs, HOV-SG, DAAAM, Clio, and active approaches such as Active Semantic Perception, SCOUT, and RGB-only Active 3D Scene Graph Generation, distinguishing passive mapping, incremental graph construction, downstream navigation, and genuine active mapping.

- Analyze existing active 3D scene-graph methods at source-code level, focusing on closed-loop systems in which the current graph guides the next viewpoint, frontier, or sensing action and new observations update the graph.
    - Trace and evaluate the complete perception–mapping–planning loop, from sensor input and geometric reconstruction through semantic node/edge inference, hierarchical graph updates, viewpoint generation, information-gain or uncertainty-based selection, motion planning, and map revision.
    - Assess scene-graph accuracy, semantic and geometric coverage, uncertainty reduction, exploration time, path length, and discovery rate. Conduct code-level ablations comparing semantic graph-guided planning with geometry-only frontier exploration and flat-map alternatives.

- Reproduce selected active-mapping baselines in simulation and visualize graph evolution, candidate viewpoints, information gain, robot trajectories, and mapping outputs using system diagrams and RViz.

Task 2: Active Mapping Capability Evaluation

2.1 Mapping in Atypical Environments

- Evaluate active 3D scene-graph mapping in cluttered, irregular, and semantically ambiguous indoor environments.

- Measure object and room recall, graph consistency, semantic coverage, and robustness to incomplete observations.

2.2 Task-Relevant Scene-Graph Evaluation

- Test whether the actively constructed hierarchical scene graph supports downstream queries such as object localization, room identification, and target-directed navigation.
- Relate task performance to graph completeness and semantic accuracy rather than evaluating general human–robot interaction.

Task 3: Explore Possible Targeted Lightweight Improvements

3.1 Improving Next-Best-View Active Mapping

- Investigate possible improvements to existing NBV methods for 3D scene-graph mapping, including alternative formulations of information gain, semantic uncertainty, geometric coverage, viewpoint diversity, travel cost, and graph completeness.
- Compare and evaluate different NBV objectives and weighting strategies under identical simulation settings, including geometry-only, semantic-aware, uncertainty-aware, and hybrid planners.

3.2 Sensor-Noise Robustness

- Analyze RGB-D and pose noise caused by B2 locomotion.
- Evaluate lightweight temporal or IMU-aligned filtering and measure its effect on scene-graph accuracy and active-exploration performance.

Task 4: Real-Robot Validation

- Deploy the complete scene-graph and NBV pipeline on TIAGo and B2 robots and validate real-time sensor processing and viewpoint selection.
- Run real-world active-mapping experiments in indoor environments using RGB-D data and robot pose information.
- Compare real-robot and simulation results using scene-graph accuracy, map coverage, exploration time, trajectory quality, and computational cost.
- Evaluate system robustness under real sensor noise, tracking errors, and incomplete observations.

Task 5: Project Consolidation

- Produce a technical report covering active 3D scene-graph mapping methods, system analysis, experiments, ablations, and open challenges.
- Organize the implementation with reproducible configuration files, evaluation scripts, and README instructions.
- Deliver trajectory visualizations, scene-graph evolution plots, metric comparisons, and a final project presentation.

4. Research hypothesis of UREx project

H1: Compared with fixed-trajectory and geometry-only frontier methods, hierarchical scene-graph-guided NBV planning achieves higher scene-graph accuracy, semantic and geometric coverage, uncertainty reduction, and object/room discovery rates under equivalent exploration budgets.

H2: The resulting actively constructed scene graphs provide more reliable support for high-level tasks such as semantic queries, object localization, and goal-directed navigation, particularly in cluttered or atypical environments.